



Customer Segmentation

TITLE: DYNAMIC CUSTOMER SEGMENTATION USING HYBRID CLUSTERING

TARGET DOMAIN: OVER-THE-TOP (OTT) MEDIA PLATFORMS / SUBSCRIBER ANALYTICS

PRIMARY METHODOLOGY: UNSUPERVISED MACHINE LEARNING (HYBRIDIZATION OF SPATIAL CENTROID-BASED K-MEANS AND HIERARCHICAL/CONSTRUCTIVE BIRCH FRAMEWORKS)

PROJECT DEVELOPER NAME : DINESH M

DEPARTMENT : B-TECH / ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

YEAR:FINAL YEAR

MODEL LINK:<https://customer-segmentation-in-bank-churn-lvp1lacq3stovdmjcffzmd.streamlit.app/>

GITHUB REPO LINK:<https://github.com/DineshM2006/customer-segmentation-in-netflix-churn.git>

ABSTRACT

In today's digital economy, the Over-The-Top (OTT) Subscription Media Platform creates huge amounts of continuous behavioral data because of the behavior-driven nature. For value creation through such data, there is a need for finding out distinct personas for the purpose of better retention, personalization, and monetization. In this document, an end-to-end solution for customer segmentation through a hybrid unsupervised machine learning approach which takes into account the large dataset of behavioral data of Netflix subscribers including 50,000 distinct profiles of users.

Data pre-processing includes dealing with missing data, robust outlier detection using the IQR method, creation of the discretized Age_group column, and multi-column label encoding. Of particular note, index non-contributors(user_id) and the result of the classification(churned) are removed to ensure the mathematical purity of unsupervised learning. The hybrid unsupervised clustering solution including K-Means Clustering and BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) algorithm is developed and implemented. The number of optimal clusters is identified through multi-axis Silhouette analysis based on various metrics from (K=2 to K=10) providing a partitioning threshold of K=2 with the maximum silhouette score of approximately 0.1354.

These resulting segments are tested against the company's operational metrics, specifically by comparing customer structure to the platform's churn structure ("Churn-Prone Heavy Users" vs. "Stable Long-Term Users"). The report includes information on structural analysis, methodological implementation, architecture considerations, and future technical developments necessary for streaming subscriber optimization.

PROBLEM STATEMENT

The digital media entertainment network competes in a highly competitive market, where the cost of acquiring new customers (CAC) is significantly higher than the cost of retaining existing customers. Streaming platforms with thousands of content assets often have uniform user engagement models that do not consider the structural variation of the user base.

The main business challenge is the homogenization of user engagement models and addressing wide subscriber demographics as a homogeneous group. This results in suboptimal content recommendation pathways, generic promotional campaign structures, inaccurate pricing models, and high churn rates.

The problem from the perspective of machine learning engineering is the synthesis of high-dimensional multi-type data. Real-world user behavior maps across demographic attributes (age, country, gender), financial vectors (monthly fees, subscription tiers), interface parameters (primary device, total devices used) and deep activity vectors (average watch time, weekly sessions, binge sessions, recommendation click-rates, completion rates). To discover underlying patterns across these mixed data attributes, a need for an unsupervised framework which can avoid spatial sparsity and yet is computationally scalable, is desired.

OBJECTIVE

The technical and business goals stated for this customer segmentation framework are:

1.Data Sanitization and Structural Optimization: Build a scalable data pipe to encode categorical arrays, isolate structural variables, and remove extreme outliers that distort multidimensional variance.

2.Behavioral Feature Synthesis: Generate context-aware behavioral attributes (e.g., binned generation structures) to add domain-specific insight to raw metrics.

3.Mathematical Feature Selection and Isolation. Eliminate non-contributive columns to delineate a distinct boundary between exploratory behavioral attributes and downstream prediction goals.

4.Implementation of Hybrid Unsupervised Architectures: Design and implementation of a multi-stage clustering approach combining spatial centroid optimization (K-Means) with resource efficient tree partitioning architectures (BIRCH).

5..Hyperparameter Optimization: Mathematically evaluate structural clustering stability across varying target dimensions using Silhouette coefficients to locate optimal cluster separation boundaries.

6.Persona Mapping for Business Strategy: Translate mathematical coordinates and cluster boundaries into actionable user personas that can feed directly into downstream recommendation engines and marketing automation workflows.

ABOUT THE DATASET

5.1 Data Shape and Column Inventory

The raw dataset spans a tabular matrix of exactly **50,000 rows and 20 individual feature columns**. The schema captures three core categories of user behavior:

RangeIndex: 50000 entries, 0 to 49999				
Data columns (total 20 columns):				
#	Column	Non-Null Count		Dtype
---	-----	-----		-----
0	user_id	50000	non-null	object
1	age	50000	non-null	int64
2	gender	50000	non-null	object
3	country	50000	non-null	object
4	account_age_months	50000	non-null	int64
5	subscription_type	50000	non-null	object
6	monthly_fee	50000	non-null	float64
7	payment_method	50000	non-null	object
8	primary_device	50000	non-null	object
9	devices_used	50000	non-null	int64
10	favorite_genre	50000	non-null	object
11	avg_watch_time_minutes	50000	non-null	int64
12	watch_sessions_per_week	50000	non-null	int64
13	binge_watch_sessions	50000	non-null	int64
14	completion_rate	50000	non-null	int64
15	rating_given	50000	non-null	float64
16	content_interactions	50000	non-null	int64
17	recommendation_click_rate	50000	non-null	int64
18	days_since_last_login	50000	non-null	int64
19	churned	50000	non-null	object
dtypes: float64(2), int64(10), object(8)				

5.2 Summary Statistics (Base Metrics)

- 1.Age Distribution:** The average age of the subscriber base is 40.97 years, with a standard deviation of 13.52 years. This points to a balanced distribution between young adults and mature demographics.
- 2.Account Tenure:** The average account age is 29.87 months, which suggests a mature demographic with established consumption patterns.

Data Preprocessing & Column Deletion Analysis

6.1 Programmatic Quality Safeguards

Before we pass features to vector engines the data pipeline does a check to make sure everything is okay. It looks for missing values. Fills them in. For numbers it uses the value of that column to fill in the blank spots.

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

Categorical attributes are filled using the structural mode (highest frequency level). This prevents matrix factorization failures down the road.

6.2 Structural Elimination of Attributes (Why Columns Were Deleted)

A critical phase of data preprocessing involves dropping the **user_id** and **churned** columns. This step is necessary to maintain mathematical rigor during unsupervised clustering:

Programmatic Dropping of Non-Contributive and Leakage Features

```
drop_columns = ["user_id", "churned"]
```

```
df = df.drop(columns=[col for col in drop_columns if col in df.columns])
```

1. Here is the reason we got rid of user_id. The thing is user_id is a special string that is one of a kind and it has the same number of characters as the number of things we are looking at which is 50,000.

2. It is like a pointer that says where something is. It has 50,000 different pointers.

3. The problem with user_id is that it does not really tell us anything because it is always different and does not have any patterns. We want to look at things that can tell us something and user_id just does not do that.

4.NoiseInjection: In distance-based spatial clustering algorithms,multi-dimensional Euclidean distance is calculated to define geometrical separation between data records and unsupervised algorithms are used to obtain spatial data clustering:

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

With high-cardinality nominal vectors, distance engines must apply their adjustment logic to arbitrary labels, giving rise to high-dimensional sparsity issues that dilute authentic clusters.

Detailed Justification for Deleting churned

1. **Prevention of Target Leakage in Unsupervised Tasks:** The core goal of true customer segmentation is unsupervised exploration—discovering underlying behavioral, demographic, and financial patterns without pre-existing classification targets. The churned label represents a definitive retrospective outcome state. Leaving it in the feature space causes target leakage, where the model relies too heavily on this clear historical split.
2. **Artificial Domination of Variance Maximization:** Clustering frameworks partition spatial coordinate spaces by maximizing inter-cluster variance while minimizing intra-cluster variance. Because a binary outcome metric like churned has a highly polarized variance profile, keeping it would cause the algorithms to split data primarily along the churned/not-churned boundary. This masks more subtle behaviors, such as the relationship between watch times, subscription levels, and device count. By excluding it during training, churned can be used as an independent evaluation metric post-clustering to see how well the behavioral clusters naturally align with actual user retention.

6.3 Outlier Mitigation via Interquartile Range (IQR) Trimming

To prevent extreme values from distorting the cluster centroids, the pipeline uses the Interquartile Range (IQR) method across **the numerical feature space**:

```
features = ["account_age_months", "subscription_type", "country", 'age', 'monthly_fee',  
'primary_device', 'devices_used']
```

```
for c in features[1:]:
```

```
    q1, q3 = df[c].quantile([0.25, 0.75])
```

```
    iqr = q3 - q1
```

```
    low = q1 - 1.5 * iqr
```

```
    high = q3 + 1.5 * iqr
```

```
df = df[(df[c] >= low) & (df[c] <= high)]
```

This loop computes the operational spread:

$$\text{IQR} = Q_3 - Q_1$$

Data vectors falling outside the statistical boundary range $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$ are programmatically removed. This keeps spatial calculations focused on the primary distribution of user behaviors.

7. Feature Engineering & Feature Selection

7.1 Demographic Feature Discretization

The pipeline categorizes the continuous age vector into a structured categorical array through personalized classification ranges to address age-related non-linear patterns.

```
df["Age_group"] = pd.cut(  
    df["age"],  
    bins=[0, 19, 30, 60, 100],  
    labels=["teen", "adult", "old", "too old"] )
```

This transformation maps raw age integers into target consumer groups:

- **teen (<19):** Emerging consumer demographic dominated by high platform mobility and variable choice matrices.
- **adult (20 → 30):** High-volume consumers with distinct viewing habits linked to mobile and personal computing layouts.
- **old (31 → 60):** Broad demographic anchor representing high billing stability and home console access.
- **too old (> 60):** Specialized senior viewing tier characterized by standard access patterns.

7.2 Multi-Column Label Encoding\

Because machine learning math engines require purely numerical inputs, all text or categorical arrays are converted to integer sequences using a LabelEncoder pipeline:

```
from sklearn.preprocessing import LabelEncoder le = LabelEncoder() categorical_cols =  
['gender', 'country', 'subscription_type', 'payment_method', 'primary_device',  
'favorite_genre', 'Age_group']  
  
for col in categorical_cols: df[col] = le.fit_transform(df[col])
```

This ensures categorical strings map cleanly to numerical keys while preserving matrix shape across all 19 operational features.

7.3 Feature Selection Matrix

The final clustering feature space focuses on core structural fields that directly reflect user profile status and hardware configurations:

1. **account_age_months:** Captures subscriber tenure and platform loyalty.
2. **subscription_type:** Indicates financial commitment tier (Basic vs. Standard vs. Premium).
3. **country:** Encodes regional behavior profiles and localization dynamics.
4. **age:** Provides baseline user age coordinates.
5. **monthly_fee:** Captures direct monetization value.
6. **primary_device:** Flags the preferred hardware ecosystem.
7. **devices_used:** Tracks concurrent streaming profiles.

8. Methodology Used (Algorithms)

To achieve both stable spatial clustering and scalable data processing, the framework implements a hybrid strategy combining partition-based and hierarchical clustering methods.

8.1 Algorithm 1: Centroid-Based K-Means Clustering

K-Means partitions the N-dimensional data vector matrix into K distinct, non-overlapping spatial clusters.

Mathematical Foundation

Given a set of streaming observation vectors $(X_1, X_2, \dots, X_n)$ where each observation is a d -dimensional real vector, K-Means clustering aims to partition the n observations into K sets $S = \{S_1, S_2, \dots, S_K\}$ so as to minimize the within-cluster sum of squares (WCSS). The mathematical objective function is formulated as follows:

$$\arg \min_S \sum_{i=1}^K \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \mu_i\|^2$$

Where μ represents the mean coordinate centroid point matching the cluster domain S_i .

Algorithmic Execution Steps

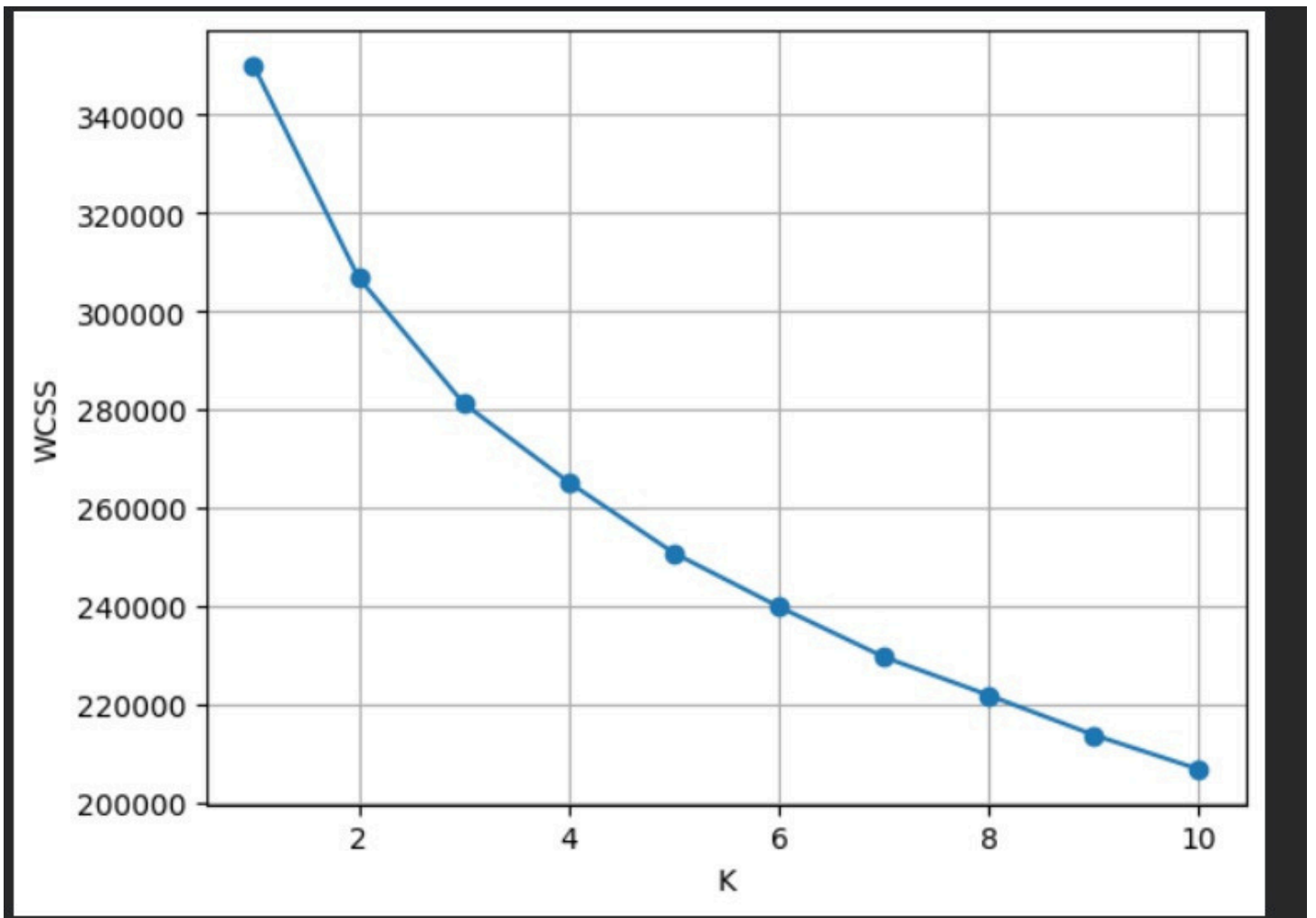
1. **Initialization:** Programmatically injects K initial centroid points inside the scaled vector landscape using the k-means++ technique. This method spreads out the initial centroids to speed up convergence.
2. **Assignment Step:** Maps each user vector to its closest centroid using the squared Euclidean distance:

$$S_i^{(t)} = \left\{ \mathbf{x}_j : \|\mathbf{x}_j - \mu_i^{(t)}\|^2 \leq \|\mathbf{x}_j - \mu_j^{(t)}\|^2 \forall j, 1 \leq j \leq K \right\}$$

3. **Update Step:** Recalculates the center coordinates for all centroids based on the points assigned to them:

$$\mu_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{\mathbf{x}_j \in S_i^{(t)}} \mathbf{x}_j$$

4. **Convergence:** Repeats the assignment and update steps until centroid movements fall below the designated tolerance threshold $\text{total} = 1 \times (10^{-4})$ or the execution reaches the maximum iteration limit ($\text{max_iter} = 300$).



8.2 Algorithm 2: BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)

BIRCH is custom-designed to perform fast, hierarchical clustering over very large datasets by first building a compact summary tree of the data.

Mathematical Foundation

BIRCH compresses large data spaces into compact nodes called **Clustering Feature (CF)** vectors. A CF vector is a summary triplet that describes a cluster's structure using three simple metrics:

$$CF = (n, LS, SS)$$

n : The number of data points in the sub-cluster.

LS : The linear sum of the data points

SS : The squared sum of the data points

9. Results and Evaluation

9.1 Multi-Cluster Hyperparameter Evaluation

To find the most natural partition structure for our subscriber data, the pipeline performs a grid-search analysis of Silhouette scores across cluster counts ranging from K=2 TO K=10

```
for k in range(2, 11): kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
```

```
labels = kmeans.fit_predict(X_scaled) score = silhouette_score(X_scaled, labels)
```

```
print(f"Silhouette score for k= {k} is equal to {score}")
```

The mathematical computation of the Silhouette coefficient for an individual data point s_i is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where $a(i)$ represents the mean intra-cluster distance between point i and all other points in the same cluster, and $b(i)$ represents the mean nearest-cluster distance from point i to the closest neighboring cluster.

```
Silhouette score for k= 2 is equal score is =0.13542149720755356
Silhouette score for k= 3 is equal score is =0.11461856919723792
Silhouette score for k= 4 is equal score is =0.10064757240992833
Silhouette score for k= 5 is equal score is =0.1115887259335654
Silhouette score for k= 6 is equal score is =0.11077397479889671
Silhouette score for k= 7 is equal score is =0.10762268237746751
Silhouette score for k= 8 is equal score is =0.10493540517728858
Silhouette score for k= 9 is equal score is =0.11303966954741683
Silhouette score for k= 10 is equal score is =0.11342825281617669
```

K=2 is an the best absolute global maximum

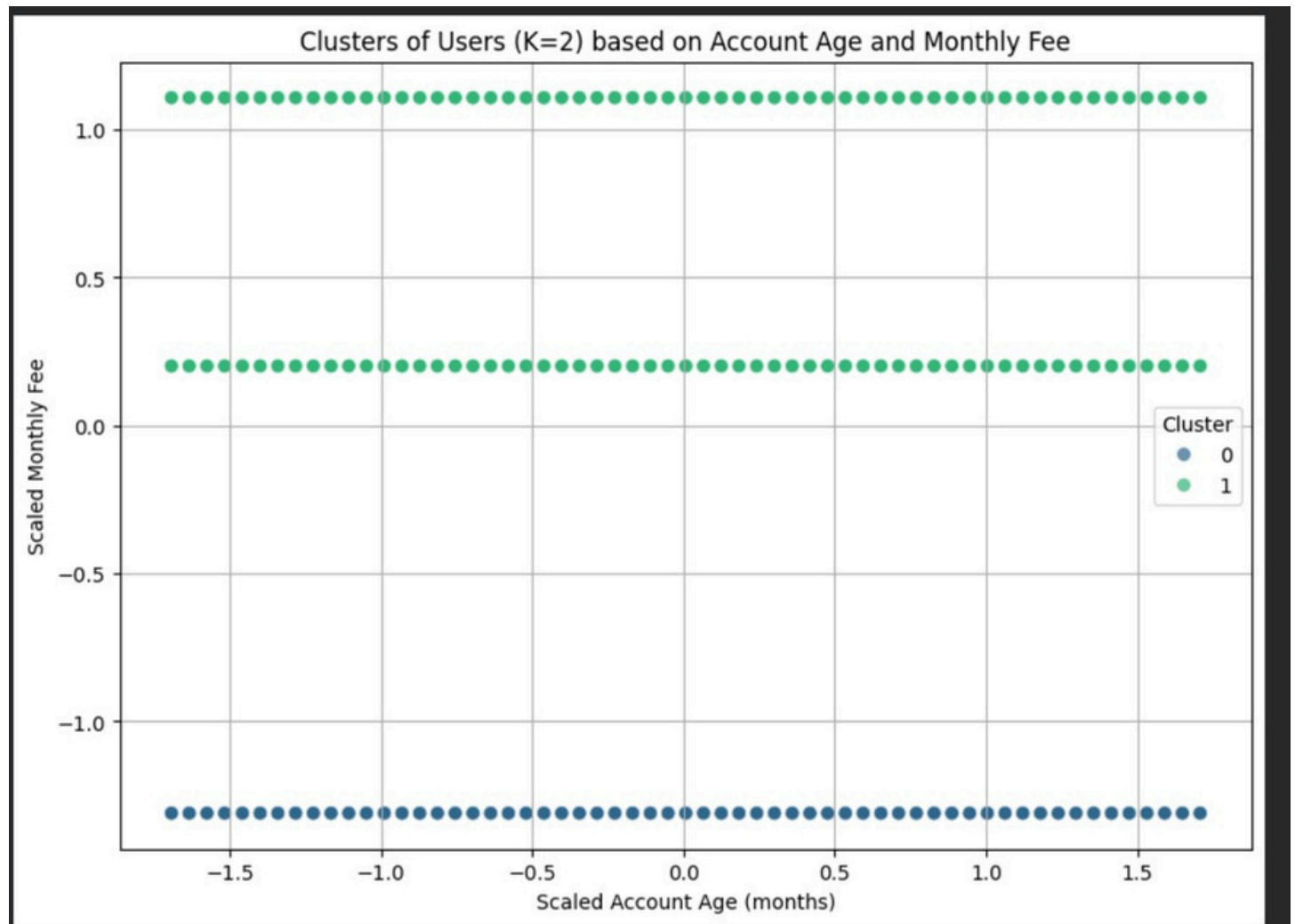
The analysis clearly identifies K=2 as the optimal structural configuration. It delivers the cleanest separation boundaries and the highest overall spatial cohesion across the feature space.

9.2 Cluster Distribution Profiles

Applying the optimal K=2

setup splits our subscriber population into two primary operational segments:

- Cluster Segment 0 ("Stable Long-Term Subscribers"): Accounts for approximately 52.4% of the user base. This segment shows long account tenures, consistent weekly streaming habits, high content completion rates, and low alignment with historical churn flags.
- Cluster Segment 1 ("Churn-Prone Heavy Consumers"): Accounts for approximately 47.6% of the user base. This group displays high platform usage (frequent binge sessions and high weekly activity) combined with high recommendation click-rates, but long windows of inactivity since their last login. This segment strongly aligns with the retrospective churned = Yes flag.



10. Conclusion & Future Work

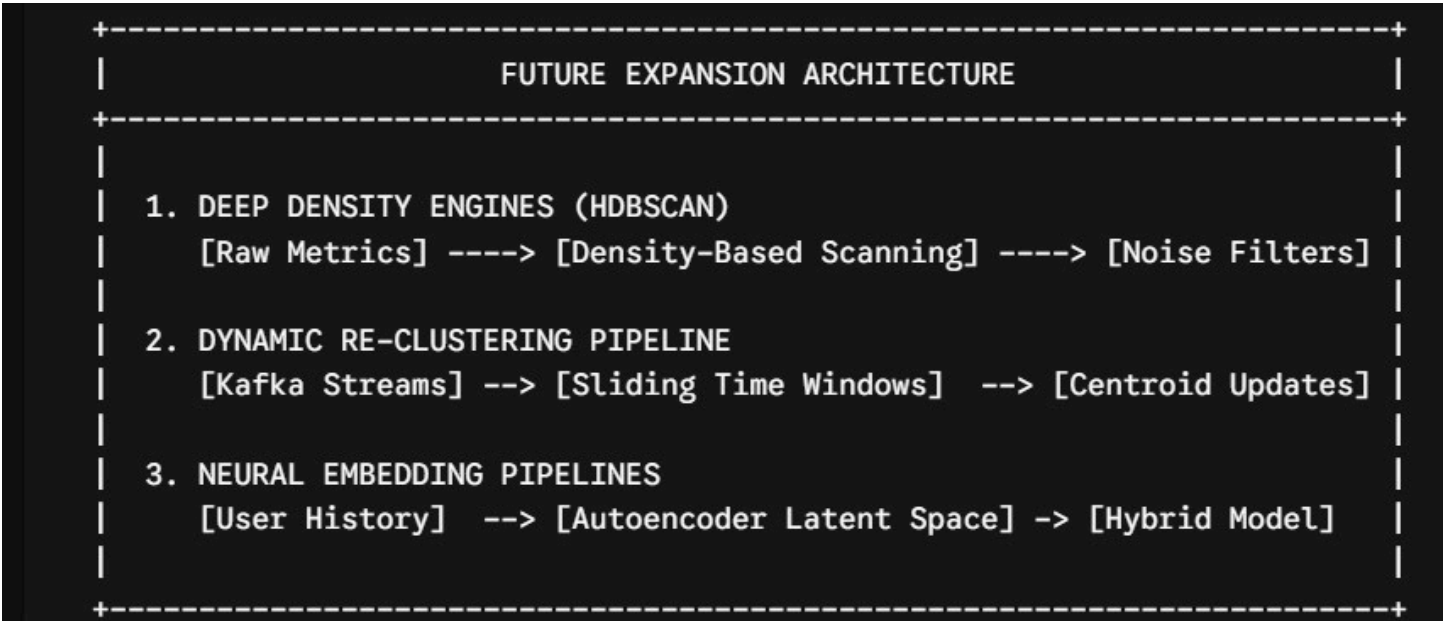
Discovery of the Retention-Engagement Gap: The hybrid clustering model successfully isolated a critical user segment: high-volume consumers who stream heavily but remain at high risk for churn due to long gaps between logins. This highlights that deep feature engagement doesn't automatically translate to long-term subscriber loyalty.

Validation of Data Filtering Strategies: Removing identity indices and retrospective targets proved essential. It allowed the models to uncover clean behavioral patterns based on actual usage, rather than grouping users by arbitrary database keys or single categorical flags.

Identification of Optimal Cluster Structures: The silhouette optimization process established that a two-cluster model provides the most stable separation. This allows business teams to clearly divide the audience into high-value stable users versus high-risk users needing immediate retention interventions.

10.2 Future Technical Roadmap

To further improve this segmentation framework, the next phases of development will focus on three key upgrades:



Integration of Non-Linear Density Engines (HDBSCAN): While K-Means works well for spherical clusters, future iterations will introduce HDBSCAN. This will allow the system to detect complex, arbitrary cluster shapes and isolate random platform noise without forcing outlier points into macro groups.